

Lesson 24 — Score & Timer Together

Matches LEGO Teacher Portal lesson: "Happy Pet".

Objective (I can...): I can use two variables to track more than one thing at once. **CS concept:** Multiple variables · **Builds on:** Lesson 23 (a variable in a decision).

AI Connection (2 min): AI juggles many numbers at once — accuracy, speed, how many examples per category. *First-principles question:* "Why can't one single number capture everything a robot needs to know?"

Materials: LEGO CS & AI kit + device per pair, trigger objects/buttons. **Model reuse:** Variables model from Lesson 21 — code only.

60-minute flow - Warm-up (8): "Two scoreboards" — pairs track points AND time with tallies, feeling the need for two boxes. - **Concept + demo (10):** Demo two variables (score and a countdown), updating each and showing both on screen. - **Build/Code (30):** Pairs program two variables — a score that counts up on an event and a timer/count that changes over a loop, both displayed. *Swap at 15 min.* - **Challenge (7):** Add an if/then that reacts when either variable hits a value (game over / win). - **Share + pack-up (5):** Pairs demo both variables changing and explain what each tracks. Model stays built.

Pair roles: Each partner owns one variable; swap at 15 min to review each other's. (*Trio: third owns the win/lose check.*)

Differentiation: *Support* — one variable pre-built; students add the second. *Extension / fast-finisher* — end the program on a combined condition. *Reteach* — build and test one variable fully, then copy the pattern for the second.

Exit check: Pair runs a program with two working, correctly labeled variables. **At-home (optional):** Design a simple game; list the two things it keeps track of. ~10 min.
